

Jiarui Li

PH.D. STUDENT · COMPUTER SCIENCE AND ENGINEERING

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Research Interests

Robotics and Embodied AI; Physical-World Robustness; Cyber-Physical Systems Security

Education

University of Michigan

PH.D. IN COMPUTER SCIENCE AND ENGINEERING

• Advisor: Prof. Z. Morley Mao

Ann Arbor, MI, USA

Sep. 2024 – Present

Carnegie Mellon University

M.S. IN ARTIFICIAL INTELLIGENCE ENGINEERING – INFORMATION SECURITY

• Advisor: Prof. Carlee Joe-Wong

Pittsburgh, PA, USA

Sep. 2022 – May 2024

The Chinese University of Hong Kong, Shenzhen

B.ENG. IN COMPUTER SCIENCE AND ENGINEERING

Shenzhen, China

Sep. 2018 – Jun. 2022

Publications

PUBLISHED

Jiarui Li, Joseph Brewington, Q. Zhang, and Z. Morley Mao. 2026. Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection. *47th IEEE Symposium on Security and Privacy (IEEE S&P)*.

Yifan Zhang, Yuren Wang, Yunta Hsieh, Xin Wang, Ping Zhang, Ziyi Yang, Jianing Ma, Zesen Zhao, Boyuan Zheng, Hei Ting Una Chan, **Jiarui Li**, Xueshen Liu, Kunxiao Gao, Ruiyao Liu, Jingxuan Zhang, Junchen Li, Zhongwei Wan, Ziheng Zhang, Jing Xiong, Shatong Zhu, Hangrui Cao, and Hui Shen. “Speculative Decoding for Multimodal Models: A Survey.” *Preprints*, 2026.

Jiarui Li, Joseph Brewington, Q. Zhang, and Z. Morley Mao. 2025. WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios. *3rd USENIX Symposium on Vehicle Security and Privacy (VehicleSec)*.

P. Sharma, **Jiarui Li**, and G. Joshi. 2024. On Improved Distributed Random Reshuffling Over Networks. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*.

T. Kim, **Jiarui Li**, N. Madaan, S. Singh, and C. Joe-Wong. 2023. Adversarial Robustness Unhardening via Backdoor Attacks in Federated Learning. *NeurIPS Workshop on Backdoors in Deep Learning*.

W. Jiao, Z. Tu, **Jiarui Li**, W. Wang, J. Tse Huang, and S. Shi. 2022. Tencent’s Multilingual Machine Translation System for WMT22 Large-Scale African Languages. *Proceedings of the Seventh Conference on Machine Translation*.

PREPRINT

Zesen Zhao, Joseph Brewington, Minkyung Cho, Ruiyang Zhu, Haoran Zhang, Hui Shen, Boyuan Zheng, **Jiarui Li**, Xueshen Liu, Fan Bai, Shuqing Zeng, and Z. Morley Mao. 2026. Synergy in Action: Benchmarking Collaborative Coordination in Multi-Arm Furniture Assembly. Submitted to the *19th European Conference on Computer Vision (ECCV)*.

Jiarui Li, Y. Yuan, and Z. Zhang. 2024. Enhancing LLM Factual Accuracy with RAG to Counter Hallucinations: A Case Study on Domain-Specific Queries in Private Knowledge Bases. *arXiv preprint*.

Research Experience

Robustness of AI-Enabled Cyber-Physical Systems

Ann Arbor, MI

ADVISOR: DR. Z. MORLEY MAO

Sep. 2024 - Present

- **Physical-World Robustness of Robotic Perception and Control** [opensource]: Designed a robotic perception robustness framework that models sensor-to-actuator dynamics, maintains a deep learning visual object tracker, and optimizes physical control inputs under motion, latency, and system uncertainty. Validated the approach using PX4/Gazebo simulation and real-world commercial UAV experiments.
- **Collaborative Robot Learning for Multi-Arm Manipulation**: Contributed to a large-scale simulation, dataset, and benchmark for evaluating VLA policies on collaborative dual-arm manipulation, spatial alignment, and long-horizon robotic assembly.
- **End-to-End Autonomous Driving Agents**: Proposed a red teaming approach to evaluate the robustness of end-to-end Vision-Language-Action (VLA) models under safety-critical edge cases.

LLM Robustness

Pittsburgh, PA

SUPERVISOR: DR. GRAHAM NEUBIG

Sep. 2023 - May. 2024

- **Retrieval-Augmented Generation** [opensource]: Designed an end-to-end RAG system for domain-specific and time-sensitive question answering over a private knowledge base, improving factual accuracy and reducing hallucinations

Machine Learning in Distributed Systems

Pittsburgh, PA

ADVISOR: DR. CARLEE JOE-WONG, DR. GAURI JOSHI

Sep. 2022 - May. 2024

- **Adversarial Federated Learning**: Developed a train-time backdoor attack in which malicious federated clients selectively undermine robustness gained through adversarial training while preserving accuracy on benign inputs.
- **Convergence Analysis**: Validated an improved convergence guarantees for distributed random-reshuffling algorithms in strongly convex and nonconvex optimization, demonstrating additional speedup as the number of networked agents increases

Multilingual Natural Language Processing

Shenzhen, Guangdong

MENTOR: DR. WENXIANG JIAO

Sep. 2021 - Sep. 2022

- **Machine Translation for Low-Resource Languages**: Contributed to a large-scale multilingual neural machine translation system covering 24 African languages and 100 evaluation directions.

Edge-to-Cloud Systems

Shenzhen, Guangdong

SUPERVISOR: DR. YEH-CHING CHUNG

Sep. 2020 - Sep. 2021

- **Real-time IoT System**: Designed an agricultural IoT monitoring system integrating embedded sensors, wireless communication, cloud infrastructure, and real-time data collection.

Presentations

2026. *Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection*. Conference Talk, 47th IEEE Symposium on Security and Privacy (IEEE S&P)..

2026. *Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection*. Invited Talk, NSF Athena AI Institute Seminar.

2026. *From Physical Signals to System Security*. Guest Lecture, Introduction to Computer Security, The University of Michigan.

2025. *WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios*. Poster, 3rd USENIX Symposium on Vehicle Security and Privacy.

Awards

- 2022 **1st Place, Constrained Track**, WMT22 Large-Scale Machine Translation
- 2021 **Bronze Medal**, "Internet+" Innovation and Entrepreneurship Competition
- 2020 **Diligentia College Scholarship**, The Chinese University of Hong Kong, Shenzhen
- 2018-19 **"Liyun" Scholarship**, The Chinese University of Hong Kong, Shenzhen

Teaching Experience

Spr 2024 **Introduction to Computer Systems**, Teaching Assistant, Carnegie Mellon University
F 2023 **Introduction to Computer Systems**, Teaching Assistant, Carnegie Mellon University
Sum 2023 **Mathematical Foundations for AI**, Teaching Assistant, Carnegie Mellon University
Spr 2023 **Adversarial Machine Learning**, Teaching Assistant, Carnegie Mellon University